

Power and Efficiency

Lesson 1–6 · the rate of doing work · $P = \Delta W / \Delta t$ · $P = F v \cos \theta$ · the efficiency of a machine

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Physics — Grade
11
 $g = 9.8 \text{ m/s}^2$

1 What is power?

Power is a measure of the **rate** of doing work, or the rate of transferring energy. It tells us *how fast* the work is done, not *how much* work is done.

$$P = \Delta W / \Delta t$$

P = average power | ΔW = work done (J) | Δt = time taken (s)

Quantity	Symbol	SI unit
work / energy	W	joule (J)
power	P	watt (W) = J / s

Remember

1 kW = 1000 W. A power of 1 W means that 1 J of work is done (or 1 J of energy is transferred) every second.

2 The power needed to raise a load

To raise a load of mass m through a height h at a steady speed, the work done against its weight is

$$W = m g h \Rightarrow P = m g h / t$$

the same load raised to the same height always needs the same work

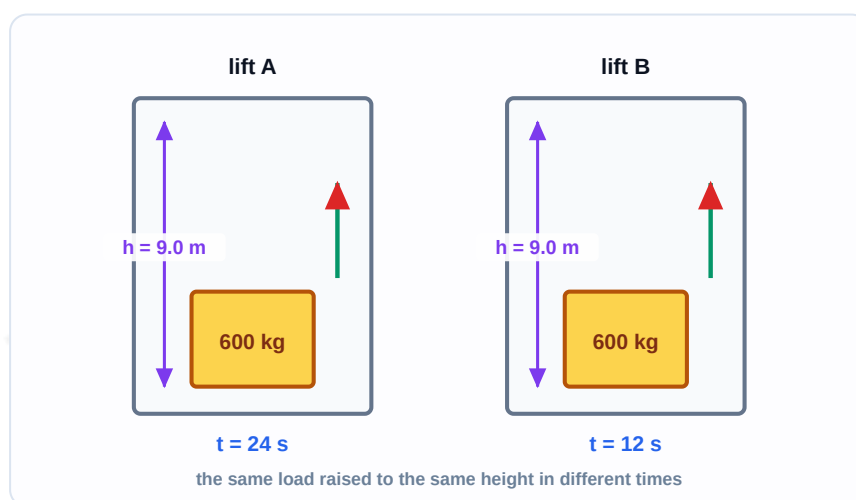


Fig. 1 — two lifts raise the same load to the same height in different times.

In Fig. 1 both lifts do the **same useful work** (same m , same h). Lift B finishes in half the time, so its power is **twice** as large. A faster machine is more **powerful**; it does not do more work.

3 The power of a force acting on a moving body

If a constant force F acts on a body moving with a velocity v , and the force makes an angle θ with the direction of motion (Fig. 2), only the component $F \cos \theta$ along the motion does work. The power of the force is

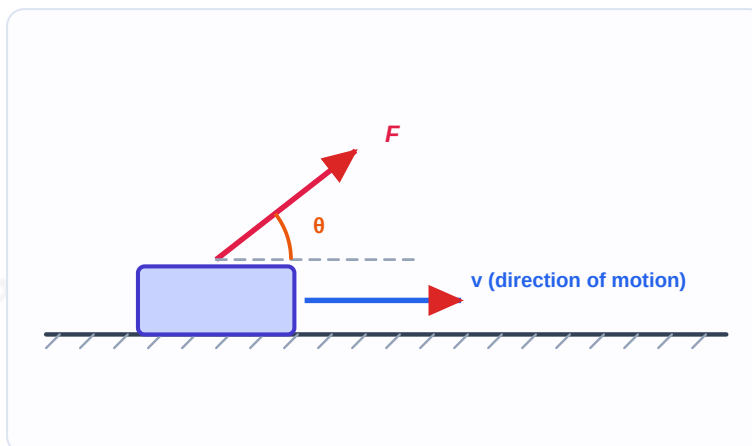


Fig. 2 — a force at an angle θ to the direction of motion.

$$P = F v \cos \theta$$

θ = the angle between the force and the direction of motion

θ	0°	37°	60°	90°
P	$F v$ (largest)	$0.8 F v$	$0.5 F v$	zero

Raising a load at a constant speed

The pulling force equals the weight ($F = m g$) and acts along the motion ($\theta = 0$), so $P = m g v$.

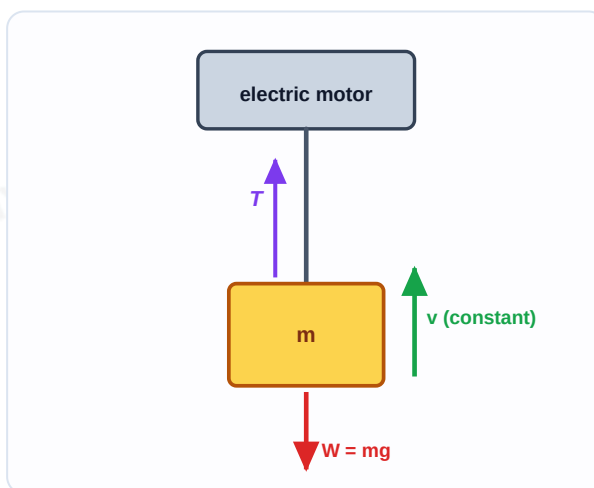


Fig. 3 — a motor raises a load at a constant speed.

4 Reading a work–time graph

On a graph of the work done W against the time t , the **slope** is $\Delta W / \Delta t$, which is the **power**. The steeper line belongs to the more powerful machine (Fig. 4).

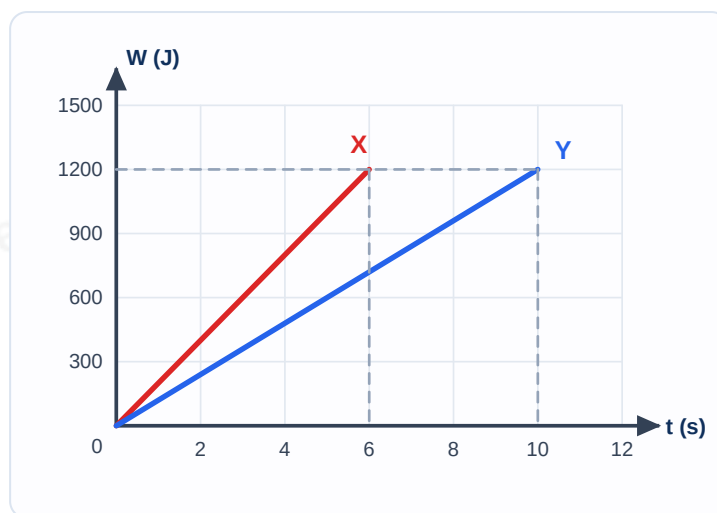


Fig. 4 — the slope of a W - t line is the power.

5 Efficiency

No real machine turns all the energy it receives into useful work. Part of the input is always wasted, mostly as heat (Fig. 5). The **efficiency** η compares the useful output with the total input:

$$\eta = (\text{useful output} / \text{total input}) \times 100 \%$$

use energies (J) or powers (W) — both give the same efficiency

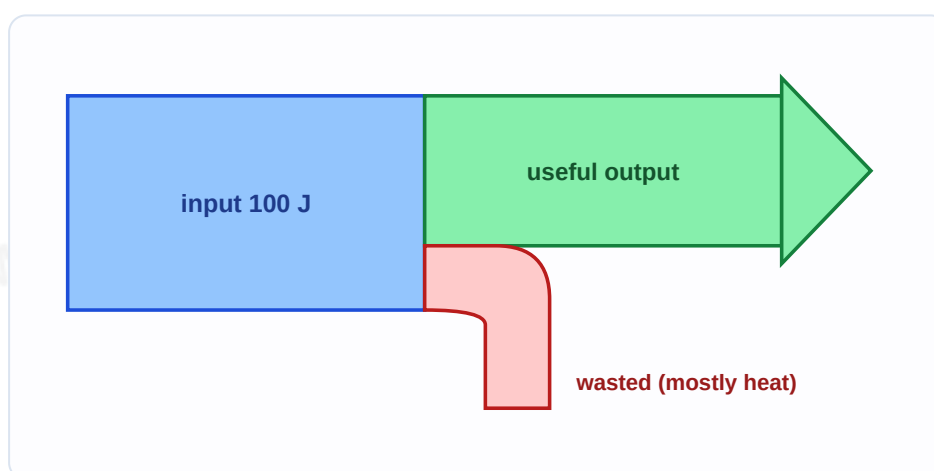


Fig. 5 — the input energy splits into a useful part and a wasted part.

- useful output = $\eta \times \text{input}$
- input = useful output / η
- wasted energy = input – useful output

An efficiency above 100 % is impossible

The useful output can never be larger than the input, because energy cannot be created. Any answer above 100 % means a mistake.

★ Key facts you must memorise

- Power = the rate of doing work or of transferring energy : $P = \Delta W / \Delta t$.
- Unit : **watt (W)** = J/s ; 1 kW = 1000 W.
- Raising a load : $W = m g h$; at a constant speed $P = m g v$.

- A force on a moving body : $P = F v \cos \theta$; it is zero when $\theta = 90^\circ$.
- Efficiency : $\eta = \text{useful output} / \text{input}$ (energies or powers), always less than 100 %.
- The slope of a W–t graph is the power.

Common mistakes

- Thinking that a more powerful machine does more work — it does the same work faster.
- Leaving the time in minutes : 2 min = 120 s.
- Using the mass instead of the weight : $W = m g h$, not $m h$.
- Using $\sin \theta$ instead of $\cos \theta$ in $P = F v \cos \theta$.
- Writing the efficiency as input / output (it would be more than 100 %).

6 Solved examples

Example 1 — two lifts

Given: Lift A and lift B each raise a load of 600 kg through 9.0 m (Fig. 1). Lift A takes 24 s and lift B takes 12 s. Find the useful work done by each lift and the power of each.

Solution

useful work (the same for both) : $W = m g h = 600 \times 9.8 \times 9.0 = 52\,920 \text{ J}$

lift A : $P = W / t = 52\,920 / 24 = 2205 \text{ W}$

lift B : $P = W / t = 52\,920 / 12 = 4410 \text{ W}$

Same work, but lift B has **double** the power because it takes half the time.

$W = 52\,920 \text{ J}$ for both ; $P(A) = 2205 \text{ W}$, $P(B) = 4410 \text{ W}$

Example 2 — a force at an angle

Given: A force of 50 N pulls a box that moves at a constant speed of 4 m/s. The force makes 60° with the direction of motion. Find the power of the force. ($\cos 60^\circ = 0.5$)

Solution

$P = F v \cos \theta = 50 \times 4 \times \cos 60^\circ = 50 \times 4 \times 0.5 = 100 \text{ W}$

If the same force acted along the motion ($\theta = 0$), the power would be $50 \times 4 = 200 \text{ W}$.

$P = 100 \text{ W}$

Example 3 — efficiency with energies



Fig. 6 — a machine of efficiency 40 %.

Given: A machine has an efficiency of 40 %. The total input energy is 250 J. Find the useful output energy and the wasted energy.

Solution

$$\text{useful output} = \eta \times \text{input} = 0.40 \times 250 = 100 \text{ J}$$

$$\text{wasted energy} = \text{input} - \text{useful} = 250 - 100 = 150 \text{ J}$$

$$\text{useful} = 100 \text{ J} , \text{ wasted} = 150 \text{ J}$$

Example 4 — efficiency with powers

Given: An electric water heater receives an electric power of 2400 W and produces a useful thermal power of 2040 W. Find its efficiency and the energy wasted in 10 minutes.

Solution

$$\eta = \text{useful power} / \text{input power} = 2040 / 2400 = 0.85 = 85 \%$$

$$\text{wasted power} = 2400 - 2040 = 360 \text{ W}$$

$$\text{wasted energy in 10 min} : 360 \times (10 \times 60) = 216\,000 \text{ J}$$

$$\eta = 85 \% , \text{ wasted energy} = 2.16 \times 10^5 \text{ J}$$

Example 5 — a motor raising a load at a constant speed

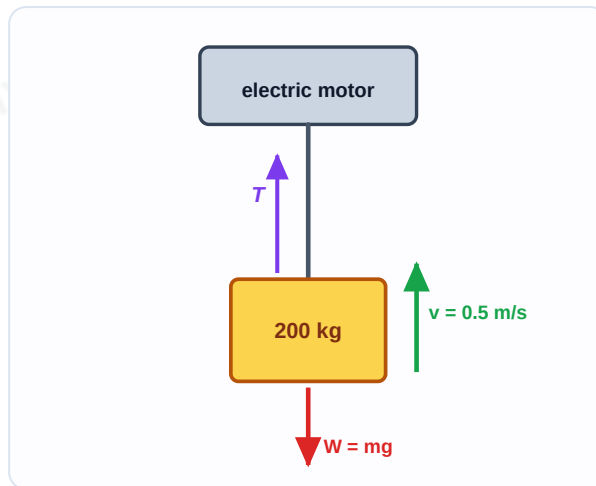


Fig. 7 — the motor takes 1400 W from the mains.

Given: An electric motor raises a load of 200 kg at a constant speed of 0.5 m/s. The motor takes an electric power of 1400 W. Find the useful power and the efficiency of the motor.

Solution

constant speed $\Rightarrow$ the pull equals the weight : $F = m g$
useful power : $P = m g v = 200 \times 9.8 \times 0.5 = 980 \text{ W}$
 $\eta = 980 / 1400 = 0.70 = 70 \%$

P(useful) = 980 W , η = 70 %

Example 6 — comparing two machines from a graph

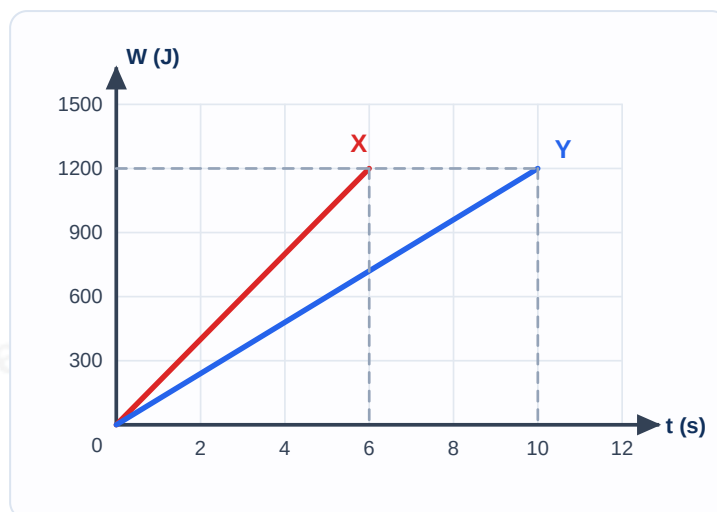


Fig. 8 — machines X and Y.

Given: Fig. 8 shows the work done by two machines X and Y against time. Find the power of each machine.

Solution

power = slope of the $W - t$ line

$$X : P = 1200 / 6 = 200 \text{ W}$$

$$Y : P = 1200 / 10 = 120 \text{ W}$$

Both machines do 1200 J, but X does it in less time, so X is more powerful.

$$P(X) = 200 \text{ W}, P(Y) = 120 \text{ W}$$

7 Exercises

22 questions — the last five are the most challenging (★). Take $g = 9.8 \text{ m/s}^2$, $\cos 37^\circ = 0.8$, $\cos 60^\circ = 0.5$. Show every step and give every answer with its unit.

Q1 Power is a measure of:

MCQ

- ☐ A) the amount of work done only
- ☐ B) the rate of doing work or of transferring energy
- ☐ C) the magnitude of the force only
- ☐ D) the total displacement of the body

Q2 The watt is equivalent to:

MCQ

- | | |
|--|--|
| ☐ A) $\text{J} \cdot \text{s}$ | ☐ B) $\text{N} \cdot \text{m}$ |
| ☐ C) J / s | ☐ D) $\text{kg} \cdot \text{m} / \text{s}$ |

Q3 A machine does 1200 J of work in 4 s. Its power is:

MCQ

- | | |
|---------------------------------|-----------------------------------|
| ☐ A) 300 W | ☐ B) 4800 W |
| ☐ C) 1196 W | ☐ D) 0.0033 W |

Q4 Two lifts raise the same load to the same height, and lift B takes half the time taken by lift A. Which statement is correct?

MCQ

- ☐ A) B does more work than A
- ☐ B) A has the greater power
- ☐ C) they do the same work, and B has double the power
- ☐ D) they have the same power, and B does more work

Q5 The efficiency of any real machine is always:

MCQ

- | | |
|--|--|
| ☐ A) exactly 100 % | ☐ B) less than 100 % |
| ☐ C) more than 100 % | ☐ D) zero |

Q6 A machine has an efficiency of 60 %. If its input energy is 500 J, the useful output energy is:

MCQ

- | | |
|--------------------------------|--------------------------------|
| ☐ A) 833 J | ☐ B) 200 J |
| ☐ C) 560 J | ☐ D) 300 J |

Q7 A lamp receives an electric power of 60 W and gives out 6 W as light. Its efficiency as a source of light is:

MCQ

A) 10 %

B) 90 %

C) 54 %

D) 1000 %

Q8 A force acts on a moving body **perpendicular** to its direction of motion. The power of this force is:

MCQ

A) Fv

B) $Fv/2$

C) zero

D) the largest possible

Q9 A crane raises a load of 500 kg through 12 m in 20 s. Find the work done and the power of the crane.

Problem

Q10 A rope pulls a box along the floor at a constant speed of 2.5 m/s with a force of 80 N that makes 37° with the direction of motion. Find the power of the force.

Problem

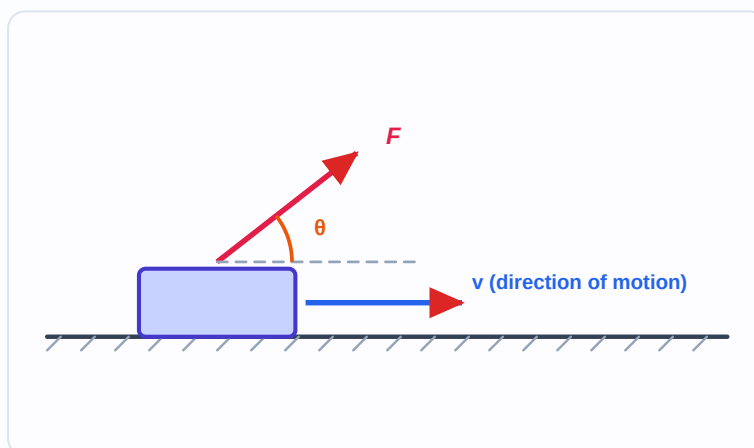


Fig. 9

Q11 An electric motor of power 1.5 kW works for 2 minutes. Find the energy it takes. If its efficiency is 80 %, find the useful work it does.

Problem

Q12 Explain why the efficiency of a machine can never be 100 % or more, and where the wasted energy goes.

Explain

Q13 Fig. 10 shows the work done by two machines X and Y against time. The more powerful machine is:

MCQ

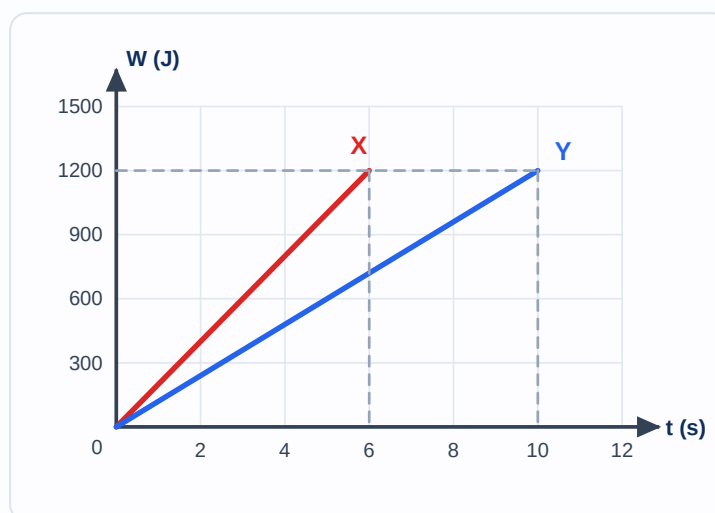


Fig. 10

A) X

B) Y

C) both have the same power

D) it cannot be known

Q14 For the machines of Fig. 10, find the power of each machine and the ratio P_X / P_Y .

Problem

Q15 An electric kettle takes a power of 2000 W and has an efficiency of 84 %. Find its useful power, and the useful and the wasted energy in 5 minutes.

Problem

Q16 A motor raises a load of mass m at a constant speed v . The useful power of the motor is:

MCQ

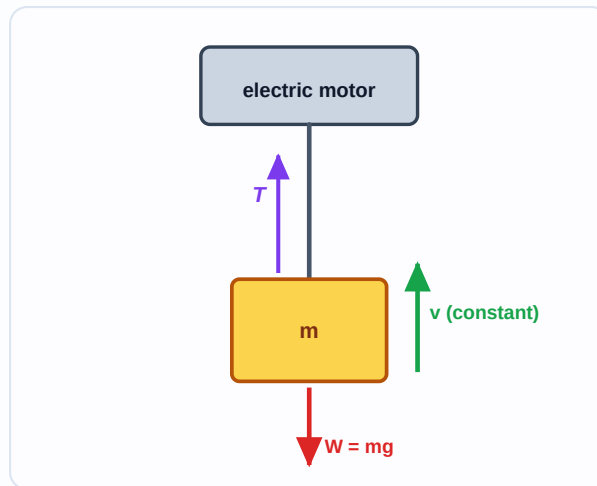


Fig. 11

A) $m v$

B) $m g v$

C) $m g / v$

D) $m g h$

Q17 A pump raises 300 kg of water through a height of 15 m every minute. Find its useful power. If the pump takes an electric power of 1000 W, find its efficiency.

Problem

Q18 A motor of efficiency 75 % raises a load of 120 kg at a constant speed of 0.5 m/s through a height of 10 m. Find: (i) the useful power, (ii) the input power, (iii) the electric energy taken by the motor during the lift. ★ HOTS

Q19 Motor A raises 80 kg through 6 m in 8 s, while motor B raises 60 kg through 10 m in 12 s. Which motor is more powerful, and what is the ratio P_A / P_B ? ★ HOTS

Q20 A force of 100 N pulls a box along the direction of motion with a power of 300 W. If the box keeps the same speed but the force now makes 60° with the direction of motion, find the speed and the new power. ★ HOTS

Q21 A heater of efficiency 80 % must give 48 000 J of useful heat every minute. Find the useful power, the input power and the energy wasted every minute. **★ HOTS**

Q22 A lift and its passengers have a total mass of 500 kg. Its motor takes an electric power of 8 kW and raises the lift through 30 m in 25 s. Find the useful work, the useful power, the efficiency of the motor and the energy wasted during the lift. **★ HOTS**

8 Model answer — step by step

Q1

power = work done / time taken : it tells how fast the work is done

✓ B) the rate of doing work or of transferring energy

Q2

$P = \Delta W / \Delta t \Rightarrow \text{unit} = \text{joule} / \text{second} \Rightarrow 1 \text{ W} = 1 \text{ J/s}$

✓ C) J / s

Q3

$P = W / t = 1200 / 4 = 300 \text{ W}$

✓ A) 300 W

Q4

same m and same $h \Rightarrow$ same work $W = m g h$

B takes half the time $\Rightarrow P(B) = W / (t/2) = 2 P(A)$

✓ C) they do the same work, and B has double the power

Q5

part of the input energy is always wasted (mostly as heat), so useful output < input

✓ B) less than 100 %

Q6

useful = $\eta \times \text{input} = 0.60 \times 500 = 300 \text{ J}$

✓ D) 300 J

Q7

$\eta = \text{useful} / \text{input} = 6 / 60 = 0.10 = 10 \%$

✓ A) 10 %

Q8

$P = F v \cos 90^\circ = F v \times 0 = 0$

(the force has no component along the motion)

✓ C) zero

Q9

$W = m g h = 500 \times 9.8 \times 12 = 58\,800 \text{ J}$

$P = W / t = 58\,800 / 20 = 2940 \text{ W}$

✓ $W = 58\,800 \text{ J}, P = 2940 \text{ W}$

Q10

$P = F v \cos \theta = 80 \times 2.5 \times \cos 37^\circ = 80 \times 2.5 \times 0.8 = 160 \text{ W}$

✓ $P = 160 \text{ W}$

Q11

$$P = 1.5 \text{ kW} = 1500 \text{ W}, \quad t = 2 \text{ min} = 120 \text{ s}$$
$$E = P t = 1500 \times 120 = 180\,000 \text{ J}$$
$$\text{useful} = \eta \times E = 0.80 \times 180\,000 = 144\,000 \text{ J}$$

$$\checkmark E = 1.8 \times 10^5 \text{ J}, \text{ useful} = 1.44 \times 10^5 \text{ J}$$

Q12

the useful output can never be larger than the input, because energy cannot be created
some of the input is always changed into heat (friction, resistance) and sound
so useful output < input $\Rightarrow \eta < 100\%$

$\checkmark$ part of the input is always lost (mostly as heat by friction) ; energy cannot be created

Q13

power = slope of the $W - t$ line ; the line of X is steeper
(X does 1200 J in 6 s, Y in 10 s)

$\checkmark$ A) X

Q14

$$P(X) = 1200 / 6 = 200 \text{ W}$$
$$P(Y) = 1200 / 10 = 120 \text{ W}$$
$$P(X) / P(Y) = 200 / 120 = 5/3$$

$$\checkmark P(X) = 200 \text{ W}, P(Y) = 120 \text{ W}, \text{ ratio} = 5/3$$

Q15

$$\text{useful power} = 0.84 \times 2000 = 1680 \text{ W}$$
$$t = 5 \text{ min} = 300 \text{ s}$$
$$\text{useful energy} = 1680 \times 300 = 504\,000 \text{ J}$$
$$\text{wasted energy} = (2000 - 1680) \times 300 = 96\,000 \text{ J}$$

$$\checkmark P = 1680 \text{ W}; \text{ useful} = 5.04 \times 10^5 \text{ J}, \text{ wasted} = 9.6 \times 10^4 \text{ J}$$

Q16

constant speed $\Rightarrow$ the pull equals the weight $m g$ and acts along the motion
 $P = F v \cos \theta^\circ = m g v$

$\checkmark$ B) $m g v$

Q17

$$\text{useful work each minute : } W = m g h = 300 \times 9.8 \times 15 = 44\,100 \text{ J}$$
$$P = W / t = 44\,100 / 60 = 735 \text{ W}$$
$$\eta = 735 / 1000 = 0.735 = 73.5\%$$

$$\checkmark P = 735 \text{ W}, \eta = 73.5\%$$

Q18

(i) $P(\text{useful}) = m g v = 120 \times 9.8 \times 0.5 = 588 \text{ W}$

(ii) $P(\text{input}) = P(\text{useful}) / \eta = 588 / 0.75 = 784 \text{ W}$

(iii) time of the lift : $t = h / v = 10 / 0.5 = 20 \text{ s}$

$$E = P(\text{input}) \times t = 784 \times 20 = 15\,680 \text{ J}$$

$$\checkmark \text{ (i) } 588 \text{ W (ii) } 784 \text{ W (iii) } 15\,680 \text{ J}$$

Q19

$$P(A) = m g h / t = 80 \times 9.8 \times 6 / 8 = 588 \text{ W}$$

$$P(B) = 60 \times 9.8 \times 10 / 12 = 490 \text{ W}$$

$$P(A) / P(B) = 588 / 490 = 1.2 \Rightarrow \text{motor A is more powerful}$$

(note : B does more work , 5880 J against 4704 J , but more slowly)

✓ A ; ratio = 1.2

Q20

$$\theta = 0 : P = F v \Rightarrow v = P / F = 300 / 100 = 3 \text{ m/s}$$

$$\theta = 60^\circ : P = F v \cos 60^\circ = 100 \times 3 \times 0.5 = 150 \text{ W}$$

✓ v = 3 m/s , P = 150 W

Q21

$$\text{useful power} = 48\,000 / 60 = 800 \text{ W}$$

$$\text{input power} = 800 / 0.80 = 1000 \text{ W}$$

$$\text{wasted energy per minute} = (1000 - 800) \times 60 = 12\,000 \text{ J}$$

✓ P(useful) = 800 W , P(input) = 1000 W , wasted = 12 000 J

Q22

$$\text{useful work} : W = m g h = 500 \times 9.8 \times 30 = 147\,000 \text{ J}$$

$$\text{useful power} : P = 147\,000 / 25 = 5880 \text{ W}$$

$$\eta = 5880 / 8000 = 0.735 = 73.5 \%$$

$$\text{input energy} = 8000 \times 25 = 200\,000 \text{ J} \Rightarrow \text{wasted} = 200\,000 - 147\,000 = 53\,000 \text{ J}$$

✓ W = 147 000 J , P = 5880 W , η = 73.5 % , wasted = 53 000 J